

Technology Stack

Date	01 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML5, CSS3, Jinja2 Templates</i>	<i>Used to create a simple, responsive, and user-friendly interface for entering applicant details and displaying prediction results.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Flask provides a lightweight web framework to process user input, load the trained machine learning model, and generate prediction results efficiently.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset & Pickle Files (model.pkl, encoders.pkl, columns.pkl)</i>	<i>The project uses CSV files for training and Pickle files to store the trained model and preprocessing objects for fast prediction.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Render</i>	<i>Render provides free, reliable cloud hosting for Flask applications with simple deployment directly from GitHub.</i>

5	Version Control & CI/CD	<i>Git, Github</i>	<i>Git is used for version control, while GitHub stores the source code and enables easy deployment and collaboration.</i>
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	<i>Scikit-learn, Pandas, NumPy, Joblib</i>	<i>Scikit-learn is used for model training, Pandas for data preprocessing, NumPy for numerical operations, and Joblib for saving/loading the trained model efficiently.</i>
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		